

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$2x + 9y = 7$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

- I. $3x + 13.5y = 10.5$
- II. $3x - 13.5y = 10.5$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables has at least one solution if the equations represent the same line or lines that intersect at exactly one point. Statement I gives the equation $3x + 13.5y = 10.5$. Multiplying both sides of the given equation, $2x + 9y = 7$, by $\frac{3}{2}$ yields $3x + 13.5y = 10.5$, so the equation in statement I is equivalent to the given equation. Therefore, the system formed by these two equations represents the same line and has infinitely many solutions. Statement II gives the equation $3x - 13.5y = 10.5$. Adding the right- and left-hand sides of $3x + 13.5y = 10.5$ and $3x - 13.5y = 10.5$ yields $3x + 13.5y + 3x - 13.5y = 10.5 + 10.5$, or $6x = 21$. Dividing both sides of this equation by 6 yields $x = \frac{21}{6}$. Therefore, the lines represented by these equations intersect at exactly one point, where $x = \frac{21}{6}$, and the system formed by the given equation and the equation in statement II has one solution. It follows that both the equation in statement I and the equation in statement II could be the other equation in a system of equations with at least one solution.

Choice A is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice B is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice D is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.